Setup for HA HDFS Cluster

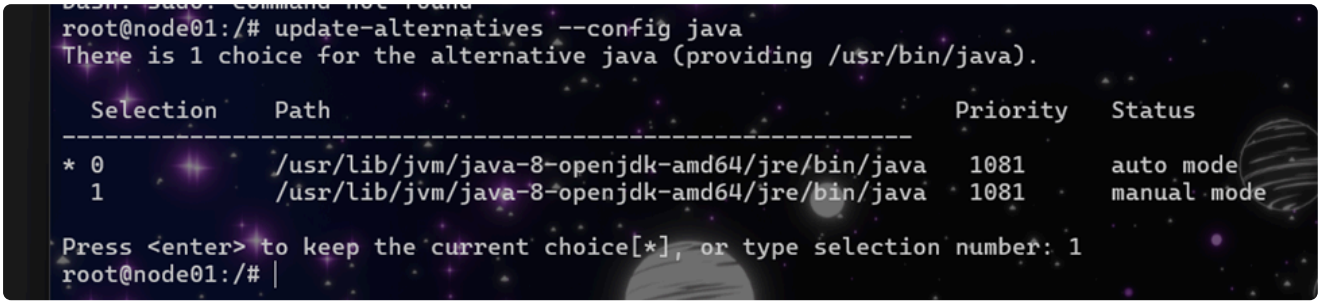
Before HDFS setup, we need to download the required prerequisites:

1. **Java JDK:**

```
sudo apt update
sudo apt install openjdk-8-jdk -y

# test the version
java -version
```

```
# set the manual or auto update of version on installing new java version
update-alternatives --config java
```



```
root@node01:/# update-alternatives --config java
There is 1 choice for the alternative java (providing /usr/bin/java).

  Selection    Path                                            Priority   Status
------------------------------------------------------------
* 0            /usr/lib/jvm/java-8-openjdk-amd64/jre/bin/java   1081      auto mode
  1            /usr/lib/jvm/java-8-openjdk-amd64/jre/bin/java   1081      manual mode

Press <enter> to keep the current choice[*], or type selection number: 1
root@node01:/#
```

```
# add java path to environment
vi /etc/environment
```

```
JAVA_HOME="/usr/lib/jvm/java-8-openjdk-amd64"
PATH="/usr/lib/jvm/java-11-openjdk-amd64/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin"
```

What is `/etc/environment`?

/etc/environment is a SYSTEM-WIDE environment file.

- It applies to ALL users on the system
- It applies to ALL processes and services
- It loads at BOOT TIME before anything else starts
- It is NOT a script — just simple KEY=VALUE pairs

```
source /etc/environment
echo $JAVA_HOME

# expected output: /usr/lib/jvm/java-8-openjdk-amd64
```

links that helped me:

- https://www.ionos.com/digitalguide/server/configuration/install-java-on-ubuntu/
- https://www.digitalocean.com/community/tutorials/how-to-install-java-with-apt-on-ubuntu-22-04

2. install ssh and pdsh

```
sudo apt-get install ssh
sudo apt-get install pdsh
```

WHY?

- automation: when you running `start-dfs.sh` you are sitting on one node, and you need to ssh to other nodes during running the script
- we also need to install sshd and must be running on every node so they listen for the NameNode's connection
- we also need it for fencing

```
# 1. Generate your key
ssh-keygen -t rsa -P '' -f ~/.ssh/id_rsa

# -t rsa → use RSA encryption algorithm
# -P '' → NO password (empty passphrase)
# -f → save key to this file


# 2. Add your key to the shared authorized_keys file
cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_keys


# 3. Fix the permissions (SSH is very strict about this)
# If permissions are too open → SSH REFUSES to work
chmod 700 ~/.ssh
chmod 600 ~/.ssh/authorized_keys
```

Note that after that we share the key with all nodes of cluster, to be able to ssh them without password.
and then restart the sshd service in all nodes:

```
service sshd restart
```

3. Installing Hadoop

```
# install hadoop mirror
wget https://archive.apache.org/dist/hadoop/core/hadoop-3.4.2/hadoop-3.4.2.tar.gz
```

```
# extract the tar file
tar -xzf hadoop-3.4.0.tar.gz
```

for configuration of HA hadoop refer to: [Configure HA Hadoop]

4. installing Zookeeper

```
# install zookeepr version compatible with hadoop version
wget https://dlcdn.apache.org/zookeeper/zookeeper-3.8.6/apache-zookeeper-3.8.6-bin.tar.gz

# extract the tar file
tar -zxvf apache-zookeeper-3.8.6-bin.tar.gz -C /
```

for configuration of HA Zookeeper refer to: [Configure HA Zookeeper]